

BIOSZEN User Manual (English)

A practical guide to run BIOSZEN from raw files to reproducible outputs.

The screenshot shows the BIOSZEN Graphics and Analysis Center interface. At the top, there is a header with the BIOSZEN logo, the text "Graphics and Analysis Center", and navigation links for "Language", "Light Theme", and "Dark Theme". Below the header, there are two tabs: "Plots and Stats" and "Growth Parameter Extraction". The "Plots and Stats" tab is active. On the left side of the main panel, there are three sections for loading data: "Load Data (.xlsx/.xls/.csv)", "Load Curves (.xlsx/.xls/.csv)", and "Load design metadata (.xlsx)". Each section has a "Browse..." button and a "No file selected" message. Below these sections are two buttons: "Instructions (download)" and "Reference input files (download)". At the bottom of the left panel, there is a "Plot setup" section with radio buttons for "Scope" (By Strain, Combined) and a dropdown for "Strain". There are also radio buttons for "Plot" (Curves, Normalized control). The main panel on the right is currently empty.

IMPORTANT: If possible, use **Platemap + Curves** mode. It gives the best support for statistics, replicate QC, and complete exports.

TIP: Keep this manual open while working. Each section includes both quick actions and deeper technical reference.

Manual Map

- [1. Before You Start](#)
- [2. Fast Start by Scenario](#)
- [3. Choose an Input Mode](#)
- [4. Input Specifications](#)
- [5. Standard Workflow](#)
- [6. Plot Types and Controls](#)
- [7. Normalization](#)
- [8. Statistics](#)
- [9. Significance Annotations](#)
- [10. QC and Replicate Management](#)
- [11. Metadata and Reproducibility](#)
- [12. Downloads](#)
- [13. Growth Module](#)

- [14. Troubleshooting Playbook](#)
- [15. Support](#)

1. Before You Start

Requirements:

- R $\geq$ 4.1.
- BIOSZEN launched from `app.R` or `BIOSZEN::run_app()`.
- Data file for **Load Data** in Excel (`.xlsx` , `.xls`) or CSV (`.csv`).
- Curves file for **Load Curves** in Excel (`.xlsx` , `.xls`) or CSV (`.csv`) when curves are not embedded in the main workbook.

Reference templates available in-app (**Reference input files (download)**) and in:

- `inst/app/www/reference_files/`

Template files:

- [Ejemplo_platemap_parametros.xlsx](#)
- [Ejemplo_curvas.xlsx](#)
- [Ejemplo_parametros_agrupados.xlsx](#)
- [Ejemplo_input_summary_mean_sd.xlsx](#)

NOTE: First launch may install packages into local `R_libs`. Keep that folder to avoid reinstalling dependencies.

2. Fast Start by Scenario

Scenario A: I have raw plate data and curves (recommended)

1. Load platemap in **Load Data**.
2. Load curves file in **Load Curves**.
3. Select scope and plot type.
4. Apply filters and replicate QC.
5. Run stats and annotations.
6. Export plot, tables, metadata, and ZIP bundle.

Scenario B: I only have grouped or summary data

1. Load grouped/summary workbook in **Load Data**.

2. Configure plots and filters.
3. Run available stats for that mode.
4. Export plots and metadata.

Scenario C: I need performance on larger datasets

1. Start with `.csv` in **Load Data**.
2. Keep selected parameters small while iterating.
3. Add overlays/advanced layers only near final export.

Plot setup
Scope:
☒ By Strain
☐ Combined
Strain:

Cepa1

Parameter:

Parameter 1

Plot:
☒ Boxplot
☐ Barplot
☐ Violin
☐ Curves
☐ Stacked
☐ Correlation
☐ Heatmap
☐ Correlation Matrix
☐ Normalize by control

3. Choose an Input Mode

- **Platemap + Curves**
Best when: You want full workflow depth.
Main limitations: Requires strict well mapping and sheet structure.
- **Grouped parameters**
Best when: Parameter-only analysis.
Main limitations: Curves require embedded `Curves_Summary` -type sheets (or a separate file in **Load Curves**).
- **Summary (Mean/SD/N)**
Best when: Raw replicate rows are unavailable.
Main limitations: Some normality/non-parametric routes may be limited.
- **CSV mode**
Best when: High-volume data and faster IO.
Main limitations: Metadata roundtrip still uses `.xlsx`.

4. Input Specifications

4.1 Platemap workbook

Required sheets:

- `Datos` : metadata + parameters.
- `PlotSettings` : default axis settings by parameter.

`Datos` expected columns:

- `Well` : well ID (`A1` , `B3` , etc.), required for curves linking.
- `Strain` : strain or biological group.
- `Media` : condition/treatment (`Control` , `Drug A` , etc.).
- `BiologicalReplicate` : biological replicate ID (`1` , `2` , `3` , ...).
- `TechnicalReplicate` : technical replicate within each biological replicate (`A` , `B` , `C` or `1` , `2` , `3`).
- `Replicate` (compatibility): legacy alternate biological replicate field.
- `Orden` : integer used for plotting/export ordering.
- Parameter columns: one or more numeric analytes/metrics.

Practical consistency rule:

- `Strain` + `Media` + `BiologicalReplicate` + `TechnicalReplicate` should identify each experimental row consistently.

`PlotSettings` expected columns:

- `Parameter`
- `Y_Max`
- `Interval`
- `Y_Title`

4.2 Curves file

Excel (`.xlsx` , `.xls`):

- `Sheet1` : first column `Time` , remaining columns by well (`A1` , `A2` , ...).
- `Sheet2` : `X_Max` , `Interval_X` , `Y_Max` , `Interval_Y` , `X_Title` , `Y_Title` .

CSV (`.csv`):

- First column `Time` , remaining columns by well (`A1` , `A2` , ...).

- Axis settings are auto-generated:
 - `X_Max` and `Y_Max` : observed maxima.
 - `Interval_X` and `Interval_Y` : `max/4` .
 - `X_Title` and `Y_Title` : blank by default.

WARNING: Curves merge failures are usually caused by inconsistent well names between platemap and curves (`Well` vs curve headers).

4.3 Grouped parameters mode

- Load grouped workbook in **Load Data**.
- Designed for parameter plots/statistics from grouped sheets (for example `Parametro_1` , `Parametro_2` , ...).
- Optional embedded curves are supported via summary-curve sheets in the same workbook.
- Keep using **Load Data** for grouped workbooks (do not upload grouped files in **Load Curves**).

4.4 Summary mode

- Load summary workbook in **Load Data**.
- BIOSZEN detects parameter summaries from any of these sheet names:
 - `Parameters_Summary`
 - `Parametros_Summary`
 - `Summary_Parameters`
 - `Resumen_Parametros`
- BIOSZEN detects embedded curve summaries from any of these sheet names:
 - `Curves_Summary`
 - `Curvas_Summary`
 - `Summary_Curves`
 - `Resumen_Curvas`
- Useful when row-level raw replicates are unavailable.
- Curves plots require either a valid **Load Curves** file or an embedded curve-summary sheet.

4.5 CSV mode

- **Load Data** accepts `.csv` and auto-detects delimiter (`,` , `;` , tab, `|`).
- BIOSZEN attempts profile conversion when CSV is not already platemap-ready.
- **Load Curves** also accepts `.csv` (`Time` + wells).

4.6 Optional AI skill for input preparation

The GitHub/source repository includes an optional AI-agent skill in the `skills/bioszen-platemap-curves/` [GitHub folder](#). To use it, give that GitHub folder URL to the corresponding AI tool or agent so it can read or acquire the skill. If the agent needs local files, download the repository ZIP from <https://github.com/bioszen/BIOSZEN/archive/refs/heads/main.zip> and copy the `skills/bioszen-platemap-curves/` folder into your agent's skill system.

Use this skill from Codex, Claude, Antigravity, or similar agentic tools when you need to generate a `Datos` + `PlotSettings` platemap from any readable data file, correct an existing platemap, repair parameter-name typing mistakes between `Datos` columns and `PlotSettings$Parameter`, prepare a separate curves workbook, or validate that `Datos$Well` exactly matches curve headers before uploading files to BIOSZEN.

The skill is a documentation/tooling extra. It does not modify the BIOSZEN app and does not assume fixed parameter names, fixed experiment labels, or a specific measurement type.

5. Standard Workflow

1. Load main data file.
2. Optionally load/merge curves.
3. Optionally load metadata.
4. Choose scope (`By Strain` or `Combined`).
5. Select plot type.
6. Apply filters and replicate selections.
7. Optionally normalize by control.
8. Run statistics.
9. Add significance annotations.
10. Export outputs.

Filters and replicates

Exclude replicate(s) in all groups

Filter Media

☒ Select / Deselect all

Media:

- ☒ Ampicillin 1µM
- ☒ Ampicillin 2µM
- ☒ Ampicillin 5µM
- ☒ Control

Replicates by Media

6. Plot Types and Controls

Boxplot

- Best for raw replicate distributions.
- Controls: jitter, box width, point size.
- Supports manual/automatic significance annotations.
- `Flip orientation (horizontal)` improves readability for long group labels.

Barplot

- Best for summarized group comparisons.
- Supports error bars and optional raw points.
- Horizontal orientation available.

Violin

- Best for showing distribution shape with replicate overlays.
- Uses the same annotation workflow as Boxplot/Barplot.
- Horizontal orientation available.

Stacked

- Parameter selector + parameter ordering controls.
- Configurable deviation bars and color behavior.
- Annotation labels/brackets available.
- Horizontal orientation available.

Correlation

- Select X/Y parameters.
- Methods: Pearson, Spearman, Kendall.
- Optional overlays: regression line, `r`, `p`, `R2`, equation.
- Advanced panel supports one-vs-all style screening and Excel export.

Heatmap

- Parameter subset selection.
- Scale options: none, row, column.
- Optional clustering and dendrograms.

- Optional in-cell value labels.

Correlation Matrix

- Multi-select parameters.
- Correlation method + p-value correction.
- Optional significant labels only.

Curves

- Configure axes, labels, line width.
- Choose line geometry and confidence interval style.
- Optional raw replicate trajectories.

Shared plot appearance controls

- The **Error bar statistic** selector controls deviation bars where available:
 - **SD** : mean +/- standard deviation.
 - **SEM** : mean +/- standard error.
 - **Min-Max** : observed minimum to maximum; available only for Boxplot.
- The collapsible **Text styling** section is available for individual plots.
- **Font family** applies to all text in the current graph. Available choices include common publication and system fonts such as Helvetica, Arial, Calibri, Cambria, Segoe UI, Times New Roman, Georgia, Verdana, and related variants.
- Bold, italic, and underline are applied independently by text part: plot title, axis titles, axis numbers, legend, data labels, and significance text.
- Each text part can use its own combination of styles; selecting underline for significance text, for example, does not force underline on the title or legend.
- These choices are applied to the plot preview and included in exported **PNG** and **PDF** files.

Text styling ^

Font family

Segoe UI ▼

Styles by text part

Plot title

☒ Bold ☐ Italic ☐ Underline

Axis titles

☒ Bold ☐ Italic ☐ Underline

Axis numbers

☐ Bold ☐ Italic ☐ Underline

Legend

☐ Bold ☐ Italic ☐ Underline

Data labels

☐ Bold ☐ Italic ☐ Underline

Significance text

☐ Bold ☐ Italic ☐ Underline

Composition Panel

Recommended steps:

1. Click **Add to panel** from individual plots.
2. Open **Composition Panel** tab.
3. Select and order plots.
4. Configure layout (rows/columns, grid, output size).
5. Configure style (legend mode, legend side, fonts, text sizes, palette).
6. Optionally add rich text and per-plot overrides.
7. Export to **PNG** , **PPTX** , **PDF** .

Composition style controls apply in parallel to all selected plots. The composition **Text styling** section mirrors the individual plot controls: the font family is applied across all plot text, while bold/italic/underline can be selected separately for titles, axes, legends, data labels, and significance text. Composition exports preserve these settings in **PNG** , **PDF** , and **PPTX** .

Label (*, **, n.s.):

*

+ Add bar

Update selected label

Remove

Clear all

Current bars

Select bars to reorder/remove

Move up

Move down

Line width:

0.8

Offset:

0.05

Bar spacing:

0.05

Text padding:

0.01

Text size (pts):

5

☐ Hide caps

7. Normalization

Enable **Normalize by control** and pick a control medium.

- BIOSZEN creates normalized columns with `_Norm` suffix.
- Correlation supports axis-specific normalization (`both` , `X only` , `Y only`).
- Fallback logic is applied when strict control pairing is not available.

8. Statistics

Main statistical tools

- Shapiro-Wilk: `stats::shapiro.test`
- Kolmogorov-Smirnov: `stats::ks.test`
- Anderson-Darling: `nortest::ad.test`
- ANOVA: `stats::aov`
- Kruskal-Wallis: `stats::kruskal.test`
- t-test routes: `rstatix::t_test` , `rstatix::pairwise_t_test`
- Wilcoxon routes: `rstatix::wilcox_test`
- Multiple-testing correction: `stats::p.adjust`

Post hoc routes by selection:

- Tukey / Games-Howell: `rstatix`
- Dunn: `rstatix::dunn_test`
- Dunnett: `DescTools::DunnettTest`
- Scheffe, Conover, Nemenyi, DSCF: `PMCMRplus`

Curve statistics (S1 - S4):

The **Curve Statistics** accordion appears for Curves plots. Select one or more methods, then click **Run curve statistics** to generate the results table.

- S1 : stats::lm + splines::ns + stats::anova
- S2 : stats::pnorm + stats::pchisq
- S3 : stats::pnorm
- S4 : gcp1yr::auc + normality-driven comparisons (stats::t.test , stats::wilcox.test , stats::aov , stats::kruskal.test)

Comparison modes:

- All vs all
- Control vs all
- Pair

P-value correction options:

- Holm
- FDR
- Bonferroni
- None

CAUTION: In Summary mode, normality may be NA and some non-parametric paths that require raw observations are disabled.

9. Significance Annotations

Manual workflow:

1. Select Group 1 and Group 2.
2. Enter label (* , ** , *** , ns , custom text).
3. Add/reorder/edit/remove annotations.

Automatic workflow:

1. Run significance tests.
2. Open auto-annotation options.
3. Choose inclusion (significant only or all).
4. Choose label mode (stars or p-value).
5. Replace or append annotations.

10. QC and Replicate Management

Use QC panels to monitor:

- Missing values.
- Outliers by group.
- Sample size and replicate coverage.

Biological replicates

- Manual include/exclude controls.
- Automatic IQR filtering.
- Keep-N reproducibility selection.

Keep-N behavior:

- Ranks replicates by distance to the group median across selected parameters.
- Keeps lowest-score (most reproducible) replicates.

Technical replicates

Available when technical replicates are valid:

- Dedicated technical QC tab.
- Group and biological-replicate selectors.
- Global select/deselect controls.
- Automatic IQR technical outlier detection.
- Technical Keep-N per subgroup.

Replicates by Media ^

Select all replicates again

Replicates - Control

☒ 1

☒ 2

☒ 3

Replicates - Ampicillin 1µM

☒ 1

☒ 2

☒ 3

Replicates - Ampicillin 2µM

☒ 1

☒ 2

☒ 3

Replicates - Ampicillin 5µM

☒ 1

☒ 2

☒ 3

11. Metadata and Reproducibility

Metadata flow:

- **Download metadata** to save current state.
- Re-import metadata in future sessions.
- Orientation flip state persists across metadata roundtrip.
- Plot typography choices, error-bar statistic, and curve-statistics method selection persist across metadata roundtrip.

Reproducibility bundle:

- Save plot versions in-session.
- Export ZIP with plot assets + metadata.
- Reopen workflows with consistent configuration.

Regression coverage includes:

- Orientation flip applied only to Boxplot/Barplot/Violin/Stacked.
- Metadata roundtrip persistence checks.
- Final builder orientation checks.

12. Downloads

Main outputs:

- Plot image (`PNG` , `PDF` , depending on chart).
- Data export.
- Metadata export.
- Statistics export.
- Bundle ZIP.
- Advanced correlation tables.
- Merged platemap/curves exports (if merge tools were used).

Plot exports preserve the active visual configuration, including font family, per-text-part bold/italic/underline choices, selected error-bar statistic, significance labels, and axis/legend settings. Composition exports preserve the same typography controls across all plots in the layout.

13. Growth Module

Growth tab file support:

- Accepted file type: Excel (.xlsx).
- Auto-detected structures:
 - Reader/Tecan-like raw layout (typically later-row data in Sheet1).
 - Processed A1 table layout (first column time, following columns as wells/curves).

Extracted parameters:

- uMax : maximum exponential-phase slope.
- max_percap_time : time window of max per-capita growth.
- doub_time : doubling time ($\ln(2) / u_{\text{Max}}$).
- lag_time : pre-exponential transition time.
- ODmax : maximum measured OD/signal.
- max_time : time at ODmax .
- AUC : area under the curve.

Typical flow:

1. Upload one or more growth files.
2. Set max time and interval.
3. Run extraction.
4. Download ZIP outputs.
5. Reuse extracted outputs in main plotting workflows.

The screenshot displays the user interface for the Growth Module. At the top, there is a file selection area with a 'Browse...' button and the text 'No file selected'. Below this, there are two input fields: 'Max time' with the value '48' and 'Interval (by)' with the value '0.5'. Underneath the input fields are three prominent action buttons: a blue button labeled 'Calculate parameters', an orange button labeled 'Stop process', and a green button labeled 'Download results' with a download icon.

14. Troubleshooting Playbook

- **Upload error**
Likely cause: Missing sheet/column names.
What to do: Validate workbook structure and exact headers.
- **No plot generated**
Likely cause: Selected parameter/group absent after filtering.
What to do: Reset filters and verify parameter availability.
- **Only Curves appears in plot type selector**
Likely cause: No valid parameter columns were parsed from the uploaded data file.
What to do: Verify grouped/summary sheet structure and parameter headers, then re-upload.
- **Normalization unavailable**
Likely cause: Missing control medium in current scope.
What to do: Confirm control group exists in active subset.
- **Stats disabled**
Likely cause: Mode/test mismatch.
What to do: Switch test or use a mode with raw-compatible data.
- **Curves merge fails**
Likely cause: Well ID mismatch.
What to do: Match platemap `Well` values to curves columns.
- **Grouped/Summary workbook loads but Curves has no data**
Likely cause: Missing embedded curves summary sheet.
What to do: Add `Curves_Summary` (or alias) to the workbook, or upload curves separately in **Load Curves**.
- **CSV not recognized**
Likely cause: Wrong delimiter or missing required headers.
What to do: Check delimiter consistency and required columns.
- **Slow performance**
Likely cause: Too many parameters/overlays at once.
What to do: Reduce active parameters and heavy layers.

15. Support

For support and bug reports: bioszenf@gmail.com